

Blue Noise Dithering for Reservoir-based Spatio-temporal Importance Resampling

Zhong Shi and Kun Xu

Abstract—Monte Carlo path tracing remains the standard approach for photorealistic image synthesis, but its high variance makes efficient rendering challenging. Reservoir-based Spatiotemporal Importance Resampling (ReSTIR) has recently emerged as a promising framework for efficiently reusing samples in real-time path tracing. However, the stochastic resampling step of ReSTIR makes it incompatible with ideal sample distributions like blue-noise.

In this work, we reinterpret the ReSTIR process through the lens of dithered sampling and establish a connection between reservoir resampling and ordered blue-noise sampling. Building on this perspective, we propose a blue-noise-driven spatiotemporal resampling scheme that preserves perceptually favorable high-frequency error distribution during reuse.

We provide a full integration of the proposed method into a real-time path tracer. Our experiments demonstrate that integrating blue noise into ReSTIR can improve the perceptual quality of low-sample-count rendered images across a range of scenes. Quantitative evaluations show reductions in perceptual error metrics with only marginal runtime overhead, maintaining efficiency suitable for real-time rendering applications.

Index Terms—rendering, sampling, blue noise, ray tracing, importance resampling

MODERN photorealistic rendering relies on Monte Carlo (MC) sampling to produce complex light effects by approximating high-dimensional light transport integrals. Although this approach is flexible and unbiased, the inherent randomness introduces error with high variance, observed as noise in rendered images. While variance can be reduced by increasing the number of samples, this approach quickly becomes computationally prohibitive and is often beyond the budget of real-time applications. Despite improved visual content with sampling, variance noise can produce implausible visual artifacts. Consequently, a broad class of methods has focused on variance reduction, either by improving sampling efficiency or by reusing information across space and time.

Reservoir-based Spatiotemporal Importance Resampling (ReSTIR) [1], [2] represents one of the most significant advances in this area. By reusing samples from spatiotemporal neighbors through reservoir-based importance resampling, ReSTIR achieves orders-of-magnitude improvements in sample efficiency, enabling real-time path tracing in industrial production [3]. Since its introduction, several extensions have been developed, such as MCMC extensions [4] and better sample reprojection [5].

Zhong Shi and Kun Xu are with the Department of Computer Science and Technology, Tsinghua University, Beijing 100084, China.

Kun Xu is the corresponding author (e-mail: xukun@tsinghua.edu.cn).

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Despite these advances, sample variance remains a major challenge. Although ReSTIR produces good samples from complex light-transport distributions induced by multiple lights, materials, and visibility, the finite candidate set and single selected representative can still leave visible residual error, especially as color noise.

An alternative perspective on noise reduction is not to reduce variance directly but to distribute it less perceptibly. Noise patterns with high-frequency spectral characteristics, such as blue noise, are known to be less visually distracting and more easily filtered by the human visual system [6]. However, integrating blue-noise sampling directly into ReSTIR is non-trivial. The repeated resampling and reservoir updates rapidly destroy the structured correlations that define blue-noise distributions in candidates or random numbers. Previous attempts [7] have only applied blue noise to simple resampling cases, which is incompatible with the ReSTIR pipeline.

In this work, we propose a blue-noise-driven spatiotemporal resampling framework that combines the perceptual advantages of blue-noise sampling with the sample efficiency of ReSTIR. Inspired by the connection between ReSTIR and dithering, we find that the weighted candidate selection can be viewed as a discrete analog of blue-noise dithering. Building on this connection, we introduce a value-guided candidate ordering strategy that maintains blue-noise error characteristics across spatiotemporal reuse. We introduce an expected linear-time algorithm for ordered resampling from unordered discrete candidates, avoiding the need for sorting.

We fully integrate our approach into a real-time path tracer, Falcor [8], and evaluate its performance across a variety of scenes. Experiments demonstrate that our method improves perceptual quality in our tests, producing more favorable residual-noise structure and lower perceptual error metrics with negligible runtime overhead. We also evaluate the method after denoising, where the reshaped residual error generally remains beneficial. Our method can be easily integrated into existing ReSTIR pipelines, providing a practical path toward perceptually guided real-time rendering.

Our main contributions are summarized as follows:

- We establish a conceptual link between the ReSTIR resampling process and dithered sampling, providing a new perspective on resampling in ReSTIR.
- We propose a blue-noise-driven spatiotemporal resampling framework that preserves the perceptually favorable error spectrum of blue noise during ReSTIR reuse.
- We demonstrate perceptual-quality improvements over standard ReSTIR across multiple scenes, supported by

quantitative metrics, visual comparisons, and denoised outputs.

I. RELATED WORKS

A. Dithering & Halftoning

Dither [9] is a classic signal-processing technique used to mask quantization artifacts by deliberately introducing noise before quantization, which helps mask banding and other structured distortions by distributing error more uniformly. As a method widely employed in both digital audio and image processing, it decorrelates the quantization error from the input signal, replacing structured artifacts with more uniform, perceptually benign noise.

In image processing, halftoning applies dithering to simulate continuous tones using a limited set of discrete color levels. Here, blue-noise masks, i.e., the patterns characterized by low power in the low-frequency band and isotropic high-frequency detail, have proven especially effective, producing visually smooth images with minimal banding and artifacts [10]. Classic dither patterns include Bayer matrix [11], void-and-cluster method [12], and their numerous extensions. Beyond procedural blue-noise generation, direct optimization methods such as error diffusion [13] and iterative halftone minimization [14] explicitly control local pixel neighborhoods to approximate ideal blue-noise characteristics.

The underlying principle is perceptual: although dithering does not reduce statistical error, it reshapes the error spectrum to align with filter methods or human sensitivity. This concept has inspired subsequent work in computer graphics, where Monte Carlo noise exhibits similar perceptual behavior.

B. Blue Noise Dithering in Rendering

Georgiev and Fajardo [15] introduced blue-noise dithered sampling in Monte Carlo rendering. Inspired by dithering, they toroidally shift each pixel's high-dimensional sample with a tiled blue-noise mask, showing minimized low-frequency error content. This results in images with higher visual fidelity at low sample counts, even though mathematical variance remained unchanged.

Subsequent research refined blue-noise generation and its application to rendering. Methods such as Filtering-Adaptive Spatiotemporal (FAST) blue noise [16] adapt the mask to specific reconstruction filters; Spatiotemporal Blue Noise Masks [17] extend blue-noise correlation across animation frames for temporal stability; and the Blue Noise Low-Discrepancy Sampler [18] improves effectiveness for sample accumulation and sample construction. Beyond these, several general blue-noise constructions, such as Z-Sobol [19], Gaussian Blue Noise [20], Optimal Transport Blue Noise [21], [22], the push-pull algorithm [23], and variational blue noise [24], have also been developed for stable performance in general integration problems. These methods highlight that variance alone does not determine perceptual quality; rather, the spatial correlation of noise plays a crucial role. Even without reducing numerical variance, properly structured blue-noise distributions can make low-sample images appear substantially cleaner.

C. Direct Blue Noise Error Optimization in Rendering

Although blue-noise sampling produces visually pleasing distributions, Monte Carlo rendering often weakens these perceptual benefits because the mapping between sample space and pixel intensity is highly complex. When the correlation between sample coordinates and pixel values is weak, blue-noise-distributed samples can still yield random, white-noise-like error.

To counter this, Heitz and Belcour [25] introduced a posterior blue-noise sampling strategy that selects candidate samples based on the local luminance histogram, ensuring that sample selection is correlated with expected signal variation. Building on this principle, Chizhov et al. [26] developed a perceptual error optimization framework that explicitly models and minimizes error visibility using the human visual system model and later extended to animation [27]. By formulating an optimization problem grounded in halftoning literature, their method produces a spatial error distribution achieving superior perceptual quality across a range of rendering conditions. These works demonstrate the possibility of applying blue-noise properties directly to the final contribution of samples.

D. Importance Resampling and ReSTIR

Importance resampling aims to efficiently approximate target distributions that cannot be sampled directly. The Resampled Importance Sampling (RIS) method [28] introduced the idea of drawing multiple candidate samples, weighting them according to importance, and resampling a single representative output.

Based on RIS, Bitterli et al. [1] developed Reservoir-based Spatio-temporal Importance Resampling (ReSTIR). Candidate selection is extended from a single pixel to spatiotemporal neighbors, exploiting sample reuse while retaining unbiasedness. By introducing reservoir sampling, the algorithm can draw a single sample from a vast number of candidates with constant storage per pixel. The method was later generalized to global illumination [29] and rigorously analyzed under the Generalized Resampled Importance Sampling (GRIS) framework [2] for path-tracing applications. Later improvements include decorrelation for local correlation patterns [4] and shading-point reprojection [5], [30], which further enhance rendering performance. Our work preserves the sampling efficiency of ReSTIR while adding blue-noise characteristics to the rendered content.

E. Combination of Resampling and Quasi-Random Sampling

In computer graphics, Ciklabakkal et al. [7] first explored this direction by combining stratified and blue-noise samples with RIS, demonstrating perceptual gains in simple resampling setups without scaling to the full ReSTIR pipeline. More recently, Salaün et al. [31] proposed a stratification strategy for ReSTIR's neighbor-candidate selection, showing reduced variance in certain scenarios. Instead of choosing neighbor candidates randomly within a spatial radius, they sort neighborhood pixels by contribution and use antithetic and blue-noise sampling to cover the sample space more

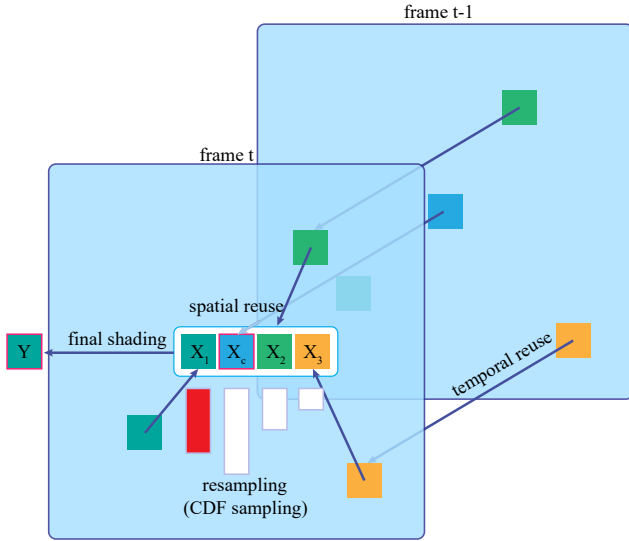


Fig. 1. Illustration of the ReSTIR process. For each frame, temporal reuse gathers candidates from the previous frame, followed by spatial reuse through resampling. The resampling step effectively performs a CDF sampling over the histogram of candidate weights.

uniformly. However, their method is built on a biased ReSTIR variant and therefore transfers less directly to an unbiased formulation, as discussed in Sec. IV-D. Their stratification operates directly on neighbor values, whereas our method injects blue-noise structure into the final resampling step after samples have been shifted and re-evaluated at the target pixel. This design preserves unbiasedness while shaping the residual error distribution more directly.

II. PRELIMINARY

A. Reservoir-based Spatio-temporal Importance Resampling

Resampled Importance Sampling (RIS) approximates sampling from a complex target distribution $\hat{p}(x)$ using independent and identically distributed (i.i.d.) samples from a simple base proposal distribution $p(x)$. Assuming we draw K candidate samples X_i ($1 \leq i \leq K$) from p , a single representative sample $Y = X_s$ is then selected from the candidate set with probability proportional to its weight w_i , defined as:

$$P(s=i) = \frac{w_i}{\sum_{j=1}^K w_j}, \quad w_i = \frac{\hat{p}(X_i)}{Kp(X_i)}. \quad (1)$$

Then for a target integral $F = \int_{\Omega} f(x)dx$, an unbiased estimator is:

$$\langle F \rangle = \frac{f(Y)}{\hat{p}(Y)} \sum_{j=1}^K w_j. \quad (2)$$

To apply RIS to the general sample-reuse setting, Generalized Resampled Importance Sampling (GRIS) [2] relaxes the requirement that samples be i.i.d. In GRIS, candidates may be generated from heterogeneous proposal domains $\{\Omega_i\}$ and may include previously resampled samples. This generalization enables efficient spatiotemporal reuse, which is crucial for rendering applications. In GRIS, instead of weighting with the reciprocal of the sample PDF, each sample carries an *unbiased contribution weight* (UCW) W_i , satisfying

$E[f(X)W_X] = \int f(x)dx$. For initial samples, this reduces to $W_i = \frac{1}{p(X_i)}$. When converting between different proposal domains, a shift mapping $T_i: \Omega_i \rightarrow \Omega$ with domain $D(T_i) \subseteq \Omega_i$ is used. The resampling weight w_i and the resulting UCW W_Y of the selected representative are then defined as:

$$w_i = \begin{cases} m_i(T_i(X_i))\hat{p}(T_i(X_i))W_i \cdot \left| \frac{\partial T_i}{\partial X_i} \right|, & X_i \in D(T_i) \\ 0, & \text{otherwise} \end{cases}, \quad (3)$$

$$W_Y = \frac{1}{\hat{p}(Y)} \sum_{j=1}^K w_j, \quad (4)$$

where m_i denotes an importance weight and $\left| \frac{\partial T_i}{\partial X_i} \right|$ is the Jacobian determinant of the shift mapping.

In rendering, GRIS leads directly to Reservoir-Based Spatiotemporal Importance Resampling (ReSTIR). Each pixel maintains a reservoir containing one candidate sample together with its UCW. For every frame, ReSTIR performs two reuse passes, as shown in Fig. 1:

- **Temporal reuse**, where the previous frame's reservoir is resampled with initial samples.
- **Spatial reuse**, where samples are drawn from neighboring pixels' reservoirs.

In practice, the intensity of the sampled rays is often used as the target distribution $\hat{p}$ in resampling. Finally, one representative sample per pixel is selected for evaluating illumination, providing efficient spatiotemporal reuse with constant memory overhead.

B. Blue Noise Dithered Sampling

Blue-noise dithered sampling aims to shift error energy toward high spatial frequencies, where it is perceptually less noticeable. Let s be a base sample shared across the image for path tracing (e.g., a low-discrepancy point or a pseudo-random sample) and let $d_i \in [0,1)^D$ denote a predefined D -dimensional blue-noise offset for pixel i . The per-pixel sample is then defined as

$$z_i = (s + d_i) \bmod 1. \quad (5)$$

z_i preserves the relative structure encoded by d_i while maintaining a uniform marginal distribution for each pixel given the shared sample s . A well-designed blue-noise mask d_i exhibits minimal energy in the low-frequency range of its spatial power spectrum while maintaining isotropic high-frequency content. Combined with the independent randomness of the base sample, this construction ensures that the estimator remains unbiased. At low sample counts, such sampling produces images where the noise appears spatially uniform and visually less disturbing than white-noise error.

However, simply injecting blue-noise offsets into the random-number stream does not guarantee improved perceptual quality, especially when the correlation between sample value and final pixel color is weak. To better exploit blue-noise properties, Heitz and Belcour [25] introduce blue noise directly into the value domain rather than the sample domain.

Let a sample u produce a scalar contribution $f(u)$, with cumulative distribution $F(x) = P(f(u) \leq x)$. A blue-noise offset d can be used to choose a contribution $F^{-1}(d)$ and further backproject to final image contribution. In practice, computing F exactly is intractable; instead, one can approximate it using a small set of neighboring or previously accumulated samples, and the blue-noise offset is applied to this empirical distribution. This perspective connects dithered sampling to sample selection: it does not merely perturb sample coordinates but instead imposes a structured, perceptually favorable distribution over sample values.

C. ReSTIR and Dithering

Although ReSTIR involves complex weighting and mapping, at its core it performs stochastic quantization by selecting one representative sample from a finite set of weighted candidates. It approximates a continuous target distribution $\hat{p}$ using a finite candidate set, from which the resampling step selects one representative. As a result, the final sample distribution becomes a mixture of neighboring candidates. From this perspective, ReSTIR shares key conceptual similarities with dithering. We highlight two complementary viewpoints that link ReSTIR to dithering:

- **Quantization viewpoint.** ReSTIR approximates a continuous probability distribution using a limited number of discrete candidates, analogous to quantization in signal processing. Dithering strategies are specifically designed to control and decorrelate quantization error, shaping its spectral characteristics to be perceptually benign. By analogy, introducing blue-noise structure into the resampling step can shape the spatial correlation of residual sampling error, pushing it to higher frequencies.
- **Stochastic filtering viewpoint.** ReSTIR weights samples in its neighborhood, which can be interpreted as a form of filtering. Although the samples are not linearly blended as in a conventional filter, the weighted resampling process still aggregates local information before selecting a representative sample. Since filtering tends to suppress high-frequency error, dithering can still be beneficial by shifting error energy toward high frequencies and making the resulting residuals less objectionable.

This analogy suggests that if the random numbers guiding ReSTIR's resampling were spatially structured—e.g., following a blue-noise distribution—the resulting sampling error could exhibit the same perceptually favorable characteristics as dithered quantization noise. When $\hat{p} = f$ for a scalar contribution, the selected representative Y does not change the scalar pixel estimate because $f(Y)W_Y = \sum_j w_j$. Therefore, structured representative selection mainly affects chromatic residuals and the stability of sample reuse rather than directly changing scalar luminance.

III. METHOD

Building upon the conceptual connection between dithering and ReSTIR established above, we now develop a practical resampling formulation that introduces blue-noise correlation into the stochastic selection process. Our objective is to shape

the spatial structure of sampling error without affecting the statistical correctness or runtime efficiency of ReSTIR. We first reinterpret ReSTIR's resampling step in a form that naturally supports blue-noise thresholds with a color metric-based ordering scheme. We then discuss the choice of the blue noise mask used and provide an efficient resampling algorithm—Quick Dithered Resample (QDR)—for practical integration. Finally, we describe how this framework is incorporated into the full ReSTIR pipeline.

A. Dithered Resampling

In ReSTIR, the resampling step selects one candidate from a discrete set of samples $\{X_i\}$ according to its cumulative weight distribution. This is equivalent to drawing a uniform random resampling variable $u \in [0, 1)$ and locating the interval in the cumulative histogram of weights where u falls.

By replacing the resampling variable with a spatially correlated blue-noise value, we can introduce structured randomness that shapes the spatial correlation of residual errors. However, naive insertion of blue-noise randoms in candidate generation quickly loses coherence through reservoir updates.

A standard weighted reservoir sampling algorithm is shown in Algorithm 1. This algorithm draws one representative sample according to the desired weighted distribution $P(s = i) = w_i / \sum_j w_j$, as required by RIS. However, because it repeatedly consumes random numbers u for independent update decisions, any spatial correlation in either the candidates X_i or the random numbers u is largely destroyed during sampling.

Algorithm 1 Weighted Reservoir Sampling

Require: Stream of candidates $\{(X_i, w_i)\}_{i=1}^K$
Ensure: Reservoir sample Y with total weight S_W

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1:  $S_W \leftarrow w_1$ 
2:  $Y \leftarrow X_1$ 
3: for  $i = 2$  to  $K$  do
4:    $S_W \leftarrow S_W + w_i$ 
5:   Draw  $u \sim \mathcal{U}[0, 1)$ 
6:   if  $u < \frac{w_i}{S_W}$  then
7:      $Y \leftarrow X_i$ 
8:   end if
9: end for
10: return  $(Y, S_W)$ 
```

For samples $\{Y_i\}$ across different pixels to yield blue-noise benefits in the final image $f(Y_i)$, they must remain as correlated with $f(Y_i)$ as possible. ReSTIR, however, depends on two distinct random streams: one for generating initial samples from the proposal distribution p , and another for resampling from reservoirs. Even if the initial samples possess blue-noise correlation, the independent resampling randomness typically eliminates it.

Heitz and Belcour [25] and Ciklabakkal et al. [7] addressed this issue with CDF sampling. The idea is to define an ordering $O(i)$ of the samples $\{X_i\}$ based on some scalar metric M , such that $M(X_{O(i)}) < M(X_{O(j)})$ for all $i < j$. Instead of making pairwise stochastic choices, we then perform inverse CDF sampling over this sorted histogram distribution. Although this

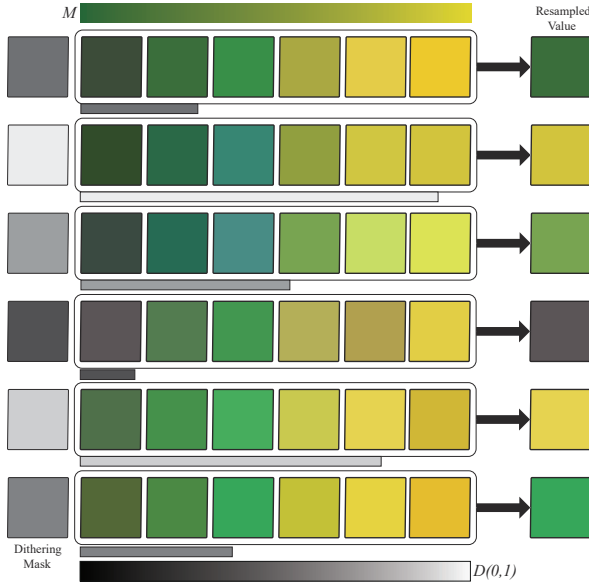


Fig. 2. By sorting candidates according to a metric M correlated with the pixel value, the resampled values can preserve blue-noise structure from the dithering mask even after resampling from different candidate sets.

procedure still performs weighted sampling from $\{X_i\}$, the random number is now positively correlated with the metric of the selected sample $M(Y)$. When the final rendering value f is correlated with M , the blue-noise property can propagate through the resampling process, as shown in Figure 2.

Previous works have explored using either Hilbert-curve order [7] or sample intensity [25] as M , but both are sub-optimal for ReSTIR. The Hilbert curve provides spatial coherence and works well in low-dimensional sampling (e.g., direct illumination). However, high-dimensional path-tracing samples often show weak correlation with the final rendering result $f(X_i)$. Moreover, the limited number of selected spatial neighbors in real-time rendering cannot fully capture such a high-dimensional structure. Using intensity as M appears intuitive but fails to preserve correlation after resampling, since ReSTIR's weight normalization diminishes the influence of intensity on the final contribution. From Eq. 4, the final contribution of the sample Y is

$$f(Y)W_Y = \frac{f(Y)}{\hat{p}(Y)} \sum_{j=1}^K w_j, \quad (6)$$

where $\hat{p}(Y)$ denotes the estimated intensity. The component varying with sample Y is therefore

$$V(Y) = \frac{f(Y)}{\hat{p}(Y)}, \quad (7)$$

that is, the RGB value normalized by intensity. To obtain a scalar ordering, we map $V(Y)$ into a single value using a 3D Morton code. Because the intensity $\hat{p}$ is computed as the luminance dot product $\hat{p}(Y) = C \cdot f(Y) = \sum_{c \in \{r,g,b\}} c_c f_c(Y)$ using the luminance-conversion vector $C = (c_r, c_g, c_b)$, each channel of $V(Y) = f(Y)/\hat{p}(Y)$ is bounded by $1/c_c$. Equivalently, the element-wise product $C \odot V(Y) \in [0, 1]^3$, where $\odot$ denotes element-wise multiplication. This bounded value can then be

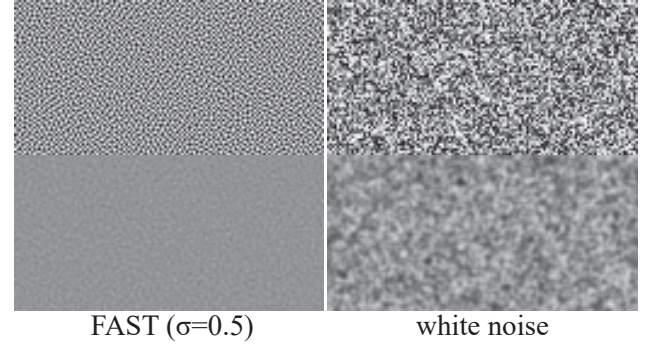


Fig. 3. Comparison of the FAST blue-noise mask we used (optimized for a Gaussian kernel with $\sigma = 0.5$) and white noise. FAST noise ensures better visual quality, as it minimizes perceptual artifacts after filtering.

directly encoded as integers for Morton ordering. This also introduces perceptual balancing across color channels, helping high-frequency residuals in any channel be transformed more uniformly into blue-noise-like error.

A natural concern is whether this added correlation affects unbiasedness. Since GRIS already accommodates correlated samples during resampling, the estimator remains unbiased in expectation. Temporal correlation in the resampling variable, however, is not explicitly covered by the original GRIS derivation. Reusing a temporally coherent dither mask can violate the practical independence assumptions behind the resampling step and may introduce bias or visible artifacts, as discussed in Sec. III-B. We therefore use a spatial blue-noise mask that is randomly translated each frame to reduce temporal correlation and mitigate this issue.

B. Choice of blue noise

Different choices of the blue-noise mask have a strong influence on visual quality. For our purposes, two properties of the mask are essential:

- Temporal decorrelation. The blue-noise mask must vary over time to avoid temporal bias and flicker.
- Spatial diversity. Neighboring pixels must exhibit sufficiently distinct samples so that spatial resampling yields diverse candidates and effective perceptual averaging.

Temporally uncorrelated sequences typically meet the first requirement. However, experiments show that statistically uncorrelated sequences can still produce visible artifacts if the blue-noise pattern repeats and creates temporary inter-frame correlations.

For example, Kronecker lattices ($d_{ij} = (ai + bj) \bmod 1$) exhibit excellent intra-frame distribution. Therefore, we may use a randomly shifted Kronecker lattice $d_{ijt} = (s_t + ai + bj) \bmod 1$ where a and b are irrational numbers and s_t is an i.i.d. sampled offset for temporal independence. This mask satisfies the second property well with fine distribution. However, when consecutive frames share similar offset s_t , the whole image shares a highly correlated dithered mask across a period of time. When pixels repeatedly choose similar samples, improper correlation may be induced.

Figure 4 shows such a case. We render a static Veach Ajar scene with a static camera, but Kronecker-lattice samples still cause significant color flickering.

Standard blue-noise masks satisfy spatial uniformity but do not explicitly optimize for perceptual filtering characteristics. To this end, we employ FAST [16], which defines blue-noise quality based on the expected error after filtering. FAST constructs noise by minimizing the following loss:

$$L = \frac{1}{N} \int D[\phi] \left(\sum_i (\Phi_i - \bar{\phi})^2 \right), \quad (8)$$

where $D[\phi]$ denotes a measure over possible integrands, and Φ and $\bar{\phi}$ denote the filtered and ground-truth integrals. This approach yields blue-noise patterns directly optimized for perceptual quality under filtering. In our method, we adopt FAST to generate the blue-noise masks. We generate a blue-noise mask of size 128×128 optimized for a Gaussian kernel with $\sigma = 0.5$, as shown in Fig. 3. For every frame, we use the randomly shifted mask u :

$$u_{i,j,t} = d_{i+r_1(t),j+r_2(t)}, \quad (9)$$

as the resample random variable, where d_{ij} is the FAST blue noise mask, and $r_i(t) \in [0, 128)$ are i.i.d. random integers used to shift the mask.

C. Quick Dithered Resample

Our dithered resampling procedure requires ordering samples by a metric M . Previous methods avoided explicit sorting per pixel by assuming ordered input or tile-based sorting. However, ReSTIR candidates are unordered, and tile-based grouping is unsuitable: it restricts sample reuse to within tiles, may cause block artifacts, and precomputed metrics based only on local information fail to capture accurate global order.

To address this, we propose the Quick Dithered Resample (QDR) algorithm. Formally, given a set of weighted samples $\{(X_i, w_i)\}$, an ordering $O(i)$ by metric M (so that $M(X_{O(i)}) < M(X_{O(j)})$ for all $i < j$), and a random number u , we select X_s such that

$$\sum_{j=1}^{s-1} w_{O(j)} \leq u \cdot \sum_{j=1}^K w_j < \sum_{j=1}^s w_{O(j)}. \quad (10)$$

This process samples from a weighted CDF over sorted candidates. The task closely resembles selecting the k -th largest element in an unordered array. If we treat each candidate as having “size” w_i , the problem can be solved using a partition-based approach. This results in the Quick Dithered Resample (QDR) algorithm. We provide pseudocode in Algorithm 2 and Algorithm 3.

Algorithm 2 Quick Dithered Resample

Require: Unordered candidates $\{(X_i, w_i)\}_{i=1}^K$, ordering metric $M(\cdot)$, resample variable $u \sim [0, 1)$

Ensure: Resampled output Y

- 1: Compute total weight $S_W \leftarrow \sum_{i=1}^K w_i$
- 2: Target position $T \leftarrow u \cdot S_W$
- 3: **return** QDR-RECURSIVE($\{(X_i, w_i)\}_{i=1}^K, M, T$)

Algorithm 3 QDR-RECURSIVE

Require: Candidate set $C = \{(X_i, w_i)\}$, metric $M(\cdot)$, target offset T

Ensure: Resampled output Y

- 1: **if** $|C| = 1$ **then**
- 2: **return** the single element of C
- 3: **end if**
- 4: Choose pivot $p \in C$ uniformly at random
- 5: Partition C into

$$L = \{ (X_i, w_i) \in C \mid M(X_i) < M(X_p) \},$$

$$R = \{ (X_i, w_i) \in C \mid M(X_i) \geq M(X_p), i \neq p \}$$

- 6: $S_L \leftarrow \sum_{i \in L} w_i$
- 7: **if** $T < S_L$ **then**
- 8: **return** QDR-RECURSIVE(L, M, T)
- 9: **else if** $T < S_L + w_p$ **then**
- 10: **return** X_p
- 11: **else**
- 12: **return** QDR-RECURSIVE($R, M, T - S_L - w_p$)
- 13: **end if**

At each iteration, we select a pivot p randomly and check whether the target sample is in the left (samples with smaller metrics) or right (samples with larger metrics). On expectation, each partition halves in size, leading to expected linear runtime. Although the algorithm is randomized, its parallel GPU implementation incurs only minor overhead. Compared to standard ReSTIR, the method requires storing all neighboring candidates locally; however, only a small set of values (radiance, sorting key, and Jacobian for path tracing) must be kept explicitly, while others can be read from reservoir storage. This overhead is small enough to fit in local memory.

D. Full integration into ReSTIR

The QDR algorithm forms the core of our dithered resampling approach. Integration into ReSTIR requires only replacing the standard resampling step with QDR and managing candidate collection explicitly.

For initial and temporal resampling passes, we retain standard Weighted Reservoir Sampling. The weight of temporal samples tends to dominate the candidate weight distribution, so dithering resampling between initial and temporal samples shows little difference with naive resampling. We therefore apply blue-noise dithering only to the spatial resampling step. Algorithm 4 summarizes the difference between standard ReSTIR and our method. Instead of resampling incrementally, we store all neighbor candidates in a local array and apply QDR once per pixel, using the blue-noise dithering mask u as the random variable that drives the selection.

For practical performance, our ReSTIR-DI implementation performs candidate reordering in sample space. To accelerate computation, the RGB radiance is simplified to intensity during the DI resampling passes. Therefore, we use sample-space ordering instead of the Morton-encoded RGB metric. Each light sample is stored as a 32-bit tag, with 2 bits for light type (environment map, analytic lights, or emissive surfaces) and 30 bits for instance index and UV coordinates. We Morton-encode

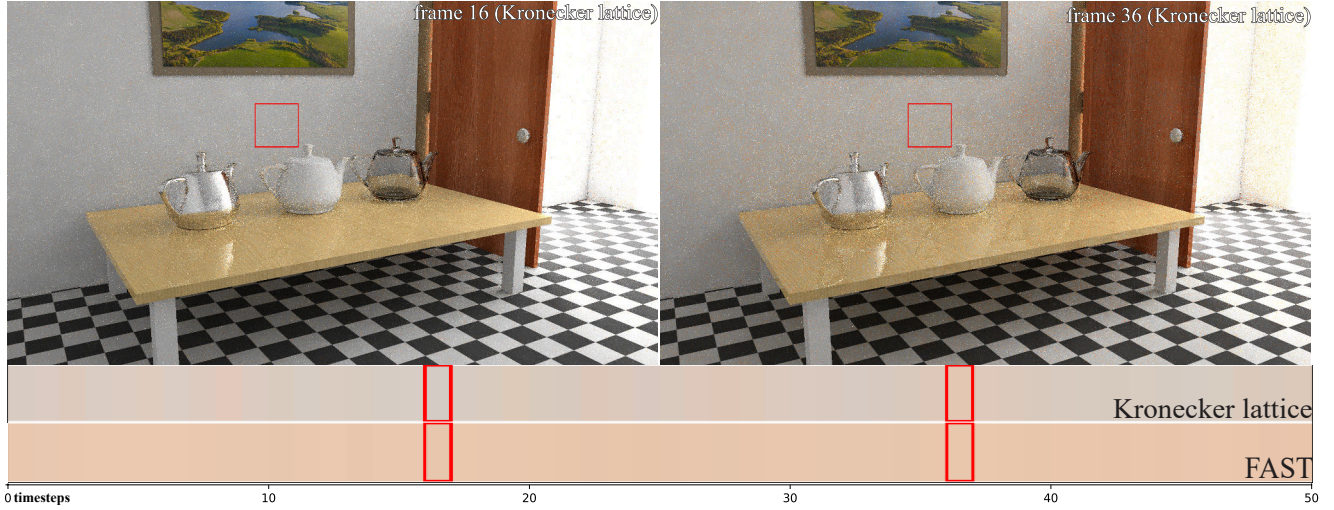


Fig. 4. An example of instability from temporal correlation. The upper images show a static scene rendered with the Kronecker lattice at different marked timesteps. The lower bars show the average color in the marked area at each timestep. With a Kronecker lattice, the color may flicker across time. With the FAST mask, the temporal correlation is weaker, reducing the flickering artifact.

the UV coordinates and then use this tag as the sorting key. Since the light sample is low-dimensional, sorting on samples can efficiently construct the correlation between the resample variable u and the final sample contribution.

Algorithm 4 Dithered spatial reuse in ReSTIR

Require: Resample neighbour set $C = \{(X_i, W_i)\}$, current sample (X_c, W)

Ensure: final sample Y

- 1: Initialize $(Y, S_W) \leftarrow (X_c, w)$
- 2: $C' \leftarrow \{(X_c, W)\}$
- 3: **for** each neighbour i **do**
- 4: Retrieve sample $T_i(X_i)$, compute weight w_i
- 5: $(Y, S_W) \leftarrow WRS(\{(T_i(X_i), w_i), (Y, S_W)\})$ (Alg. 1)
- 6: $C' \leftarrow C' + \{(T_i(X_i), w_i)\}$
- 7: **end for**
- 8: $Y \leftarrow QDR(C', M, u)$ (Alg. 2) where u from Eq. 9
- 9: **return** Y

IV. EXPERIMENT RESULTS

A. Experiment Setup

For brevity, we refer to our method as *BlueReSTIR* in this section. We compare four variants: *ReSTIR* or *ind* denotes the baseline ReSTIR implementation with independent resampling; *Ours* or *blue* denotes BlueReSTIR, i.e., ReSTIR with our blue-noise dithered resampling; *Salaiin et al.* denotes our implementation of histogram-stratified neighbor selection [31] inside the same unbiased ReSTIR framework; and *Salaiin et al. + Ours* denotes the combination of histogram-stratified neighbor selection with our blue-noise dithered resampling. We implement our method in the Falcor rendering framework [8]. For denoising results, we employ the OIDN [32] denoising library. Our method is based on ReSTIR PT [2], which also serves as the baseline. We use ReSTIR DI for direct illumination and ReSTIR PT for indirect illumination.

All parameters follow the default ReSTIR PT settings unless otherwise specified. We use a capped M of 20 for temporal weighting, sample neighbors within a radius of 20, and select 6 neighbors for reuse. All scenes are rendered with 1 sample per pixel after 200 frames of temporal accumulation.

We evaluate on several representative scenes: Bathroom, Living Room, and Fireplace Room from the Computer Graphics Archive [33]; and Bistro [34] and Sun Temple [35] from the Open Research Content Archive (ORCA). All experiments are conducted on a workstation with a Ryzen 9 9950X CPU and an NVIDIA RTX 4090 GPU, running Windows 11.

For quantitative evaluation, we report relative mean square error (relMSE) and perceptual relative mean square error (prelMSE) [26]. The prelMSE is defined as:

$$E = \frac{1}{3N} \sum_{c \in \{r, g, b\}} \left\| \frac{g * (\mathcal{T}(I'_c) - \mathcal{T}(I_c))}{g * \mathcal{T}(I_c) + \varepsilon} \right\|_2^2. \quad (11)$$

where g is a Gaussian kernel modeling the perceptual response of the human visual system, $*$ denotes convolution, and $\mathcal{T}$ denotes the tone-mapping operator applied before measuring perceptual error. The error is evaluated independently on the RGB channels $c \in \{r, g, b\}$ and then summed. Following Chizhov et al. [26], we model perceptual sensitivity using a Gaussian point-spread kernel whose standard deviation is related to display resolution and viewing distance by

$$\sigma = \frac{0.00954}{\tau}, \quad \tau = \frac{180}{\pi} 2 \arctan\left(\frac{1}{2RD}\right), \quad (12)$$

where R is the display resolution in dots per inch (dpi) and D is the viewing distance in inches. Under this model, a larger σ corresponds to a larger effective viewing distance, since a more distant observer is less sensitive to fine pixel-scale structure. In our experiments, we set $\sigma = 0.8$, close to the setting used in PE. For a 27-inch 1080p display ($R \approx 81.6$ dpi), this choice corresponds to an implied viewing distance of about 1.5m.

Our blue-noise mask is optimized for a Gaussian kernel with a smaller $\sigma = 0.5$ to compensate for randomness from

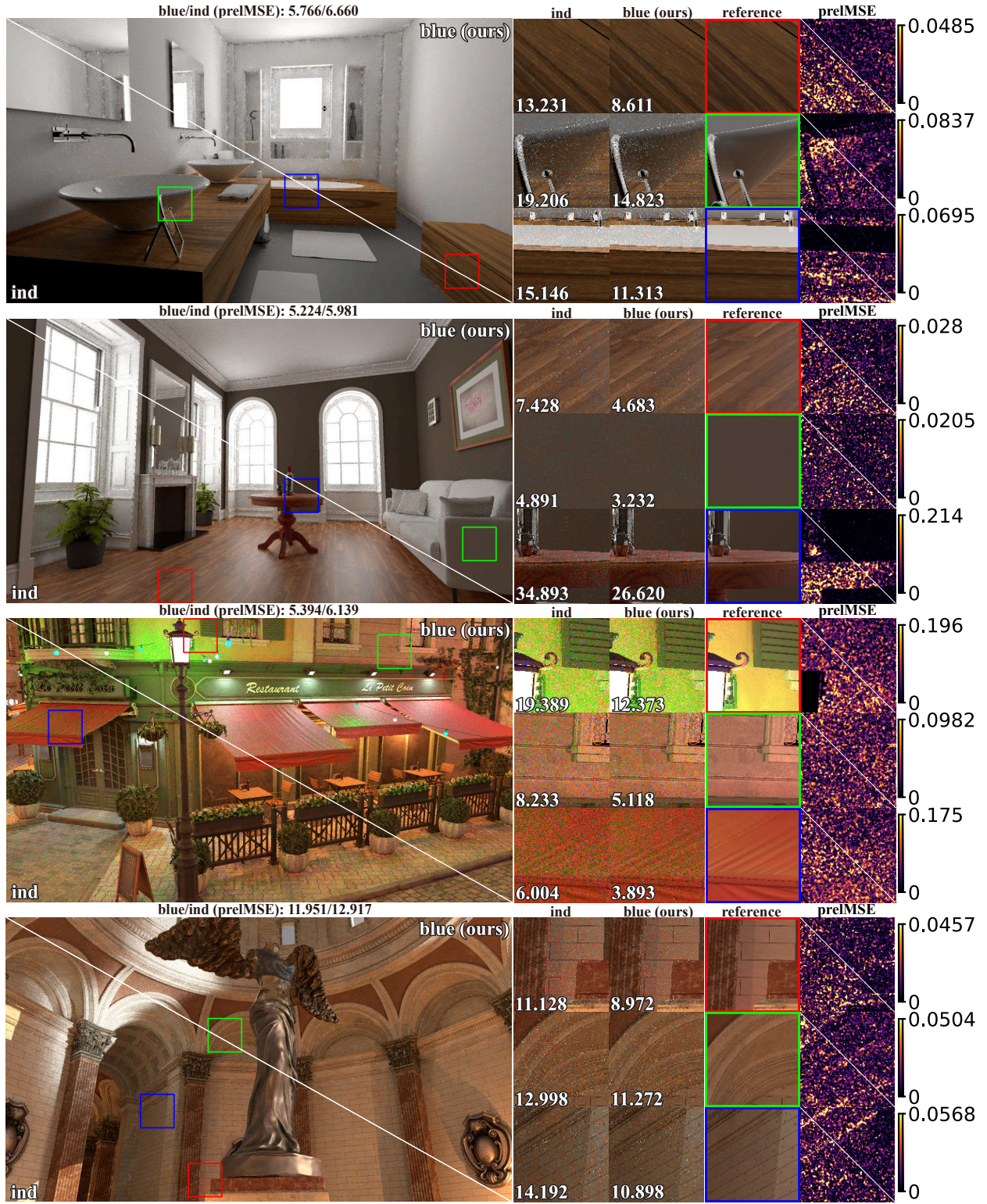


Fig. 5. Visual comparisons on the *Bathroom*, *Fireplace Room*, *Bistro*, and *Sun Temple* scenes. We compare baseline ReSTIR (ind) and BlueReSTIR (blue), together with the corresponding prelMSE ($\times 10^{-3}$) relative to the ground truth. The insets show the 1 spp rendering results and their corresponding prelMSE heatmaps. Brighter heatmap colors indicate larger local perceptual error.

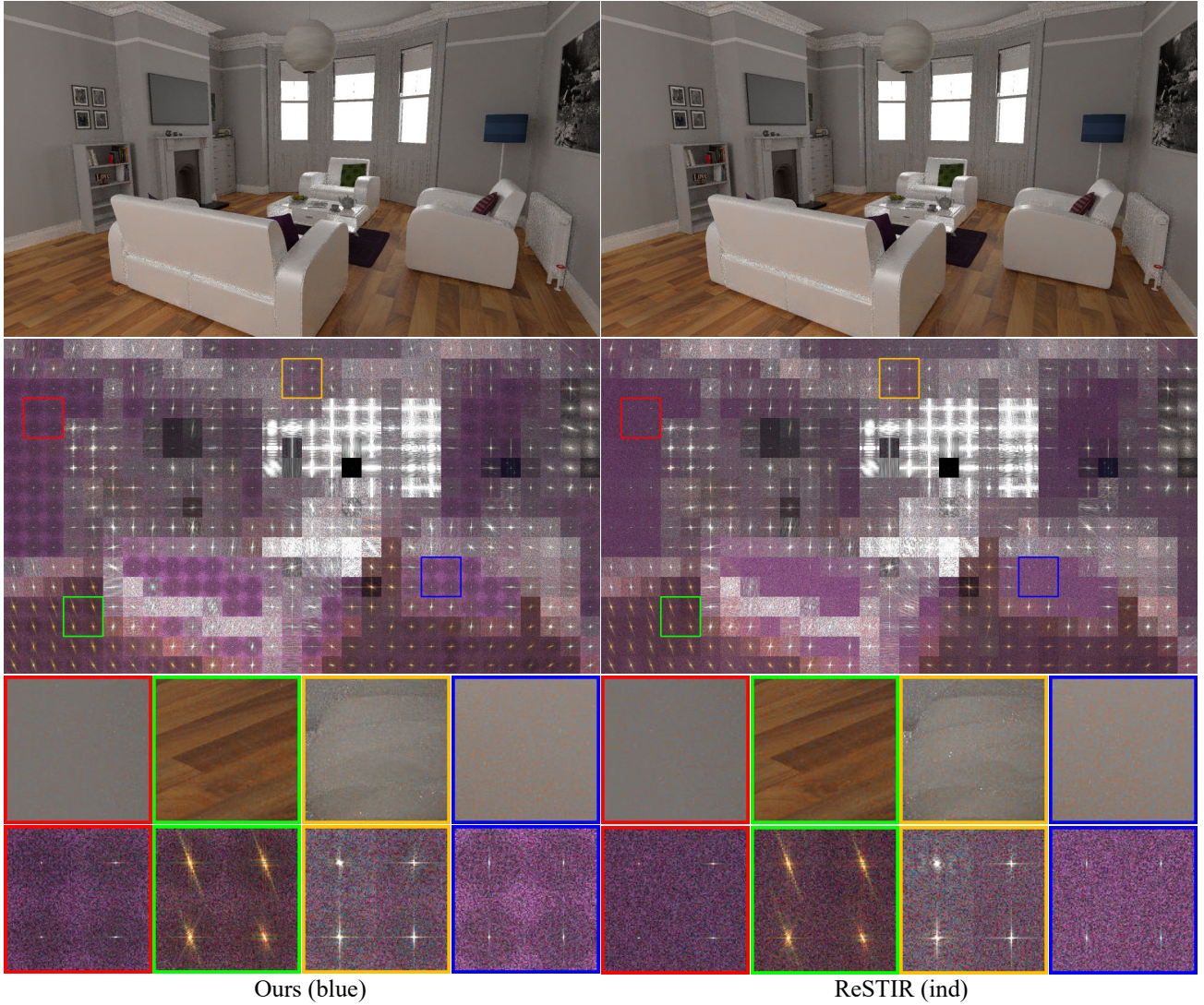


Fig. 6. Fourier spectra of the residual noise for the *Living Room* scene. Naive ReSTIR exhibits uniform (white) noise energy, while BlueReSTIR suppresses low-frequency components, shifting noise power toward higher frequencies.

other sampling processes such as neighbor selection and initial sampling.

B. Qualitative and Quantitative Results

Figure 5 presents visual comparisons across multiple test scenes. BlueReSTIR (blue) achieves lower preIMSE than the baseline independent ReSTIR (ind) in these tests and improves the perceived noise distribution especially in the highlighted regions. Because the method reshapes residual error rather than reducing estimator variance, the visual improvement is most noticeable in large smooth regions, such as walls and floors, where low-frequency blotches are more apparent.

On textured surfaces such as the wood panels in the Bathroom scene, BlueReSTIR can still reduce perceptual error because the color-based metric M is computed per pixel, allowing resampling decisions to adapt locally to surface albedo. For glossy or specular regions, such as the floor in the Fireplace Room and the awnings in the Bistro, the blue-noise

dithering can reduce coherent low-frequency visual artifacts by redistributing residual error, although isolated outliers may still persist.

In multi-light scenarios (e.g., Sun Temple and Bistro), BlueReSTIR also affects direct-illumination sampling. The dithered resampling can distribute contributions from multiple lights more evenly across neighboring pixels, reducing some locally imbalanced light-source sampling. When indirect illumination creates multimodal incoming radiance on surfaces, the RGB ordering metric can also make chromatic residuals more spatially structured.

Table I summarizes the quantitative results. The rows report relMSE, preIMSE, and denoised DSSIM for the four compared variants across all scenes. As expected, BlueReSTIR does not change relMSE significantly, since it does not reduce statistical variance directly. However, the perceptual metric (preIMSE) shows consistent improvements across all scenes, indicating that our blue-noise-driven resampling better distributes error energy under this metric. The denoised DSSIM values are also

TABLE I
QUANTITATIVE RESULTS ACROSS ALL TEST SCENES.

Metric	Method	Bathroom	Bistro	CBox	Fireplace Room	Living Room	Sun Temple
relMSE ($\times 10^{-2}$)	Ours	4.049	4.651	4.991	3.981	2.416	7.472
	ReSTIR	3.997	4.603	5.057	3.981	2.423	7.437
	Salaün et al.+Ours	4.214	4.515	5.012	3.895	2.416	7.448
	Salaün et al.	4.149	4.692	5.028	3.928	2.438	7.400
prelMSE ($\times 10^{-3}$)	Ours	5.766	5.394	6.683	5.224	3.099	11.951
	ReSTIR	6.660	6.139	8.476	5.981	3.612	12.917
	Salaün et al.+Ours	5.986	5.724	6.340	5.192	3.067	11.758
	Salaün et al.	6.848	6.264	8.048	5.912	3.559	12.530
DSSIM (denoised) ($\times 10^{-2}$)	Ours	1.817	0.073	1.018	2.517	1.499	6.193
	ReSTIR	1.926	0.074	1.562	2.669	1.658	6.143
	Salaün et al.+Ours	1.724	0.073	0.973	2.513	1.479	6.166
	Salaün et al.	1.800	0.074	1.308	2.595	1.585	6.162

TABLE II
MEAN FRAME TIME ACROSS ALL SCENES.

Scene	ind (ms)	blue (ms)
Bathroom	20.73	22.43(+8.2%)
Bistro	34.64	36.42(+5.1%)
CBox	8.65	9.21(+6.5%)
Fireplace Room	21.31	22.90(+7.5%)
Living Room	19.02	20.46(+7.6%)
Sun Temple	34.40	35.83(+4.2%)

generally improved or comparable, indicating that the reshaped error usually remains favorable after filtering.

Table II reports the average frame times for BlueReSTIR compared with baseline ReSTIR. The “ind” and “blue” columns denote the baseline and our method, respectively. Across all scenes, BlueReSTIR incurs a modest overhead between 4.2% and 8.2%, mainly due to local candidate collection and the per-pixel resampling pass. This cost remains small enough for real-time applications.

C. Spectral Analysis

To further analyze noise characteristics, we compute tiled 2D FFT spectra of residual error for the Living Room scene (Fig. 6). We apply the Fourier transform to every 64×64 tile, demonstrating the benefit of blue-noise dithered resampling from another perspective. Naive ReSTIR produces a nearly uniform white-noise spectrum, indicating uncorrelated error across all spatial frequencies. The resulting low-frequency noise forms local clusters that appear as visible artifacts. In contrast, BlueReSTIR exhibits strong attenuation of low-frequency components, with noise energy concentrated at higher frequencies. This shift aligns with the human visual system’s reduced sensitivity to high-frequency noise, which helps explain the lower prelMSE values and the smoother appearance in regions where low-frequency error is visible. The same trend is further reflected in Fig. 8, which shows the spectrum-energy distribution over spatial frequencies: the energy in the low-frequency band is effectively reduced and redistributed toward higher frequencies, rather than merely suppressed overall.

D. Comparisons with Histogram Stratification

Salaün et al. [31] proposed a histogram-based stratification strategy for ReSTIR’s neighbor-candidate selection. To compare against it fairly, we implemented their method within

Falcor’s unbiased ReSTIR framework, where reused samples are shifted and re-evaluated at the target pixel. Under this setting, histogram stratification alone provides only limited improvement over the baseline, as shown by both the visual comparisons in Fig. 7 and the spectral comparisons in Fig. 8. Since their method already introduces antithetic and blue-noise sampling during the resampling process, this comparison also suggests that direct blue-noise usage alone is not sufficient in the unbiased setting. The main reason is that histogram stratification is most effective when neighboring candidates exhibit substantial intensity variation, whereas unbiased spatiotemporal ReSTIR tends to produce more stable final candidates and values after re-evaluation and reuse. As a result, the stratification step becomes less effective, and the visual gains are generally modest without our value-guided ordering of the final target-pixel contribution. The *Bathroom* scene in Fig. 7 illustrates this local behavior. For the statue in the inset, histogram stratification can produce lower local error because its spatial masking tends to keep candidates within the same object, avoiding reuse from differently lit nearby surfaces. This is beneficial for small, coherent objects. However, this local advantage does not necessarily transfer to the full image: in regions with stronger illumination or material variation, the masking and antithetic sampling may not be as effective for gathering useful samples. BlueReSTIR instead shapes the final resampling decision after reuse, which gives more consistent aggregate prelMSE and spectral behavior in our tests.

In contrast, BlueReSTIR introduces blue-noise structure directly into the final resampling process without fundamentally modifying the ReSTIR pipeline. By shaping the spatial error distribution while preserving unbiasedness, it yields more consistent aggregate perceptual-quality improvements across scenes. When combined with histogram stratification, our resampling strategy can further improve the resulting error spectrum.

E. Ablation Study

We evaluate the effect of varying the number of spatial neighbors used during resampling. Table III compares BlueReSTIR and independent ReSTIR across three scenes (*CBox*, *Living Room*, and *Sun Temple*), reporting prelMSE for neighbor counts from $n = 3$ to $n = 6$.

Across all tested neighbor counts, BlueReSTIR consistently achieves lower prelMSE than independent ReSTIR, showing

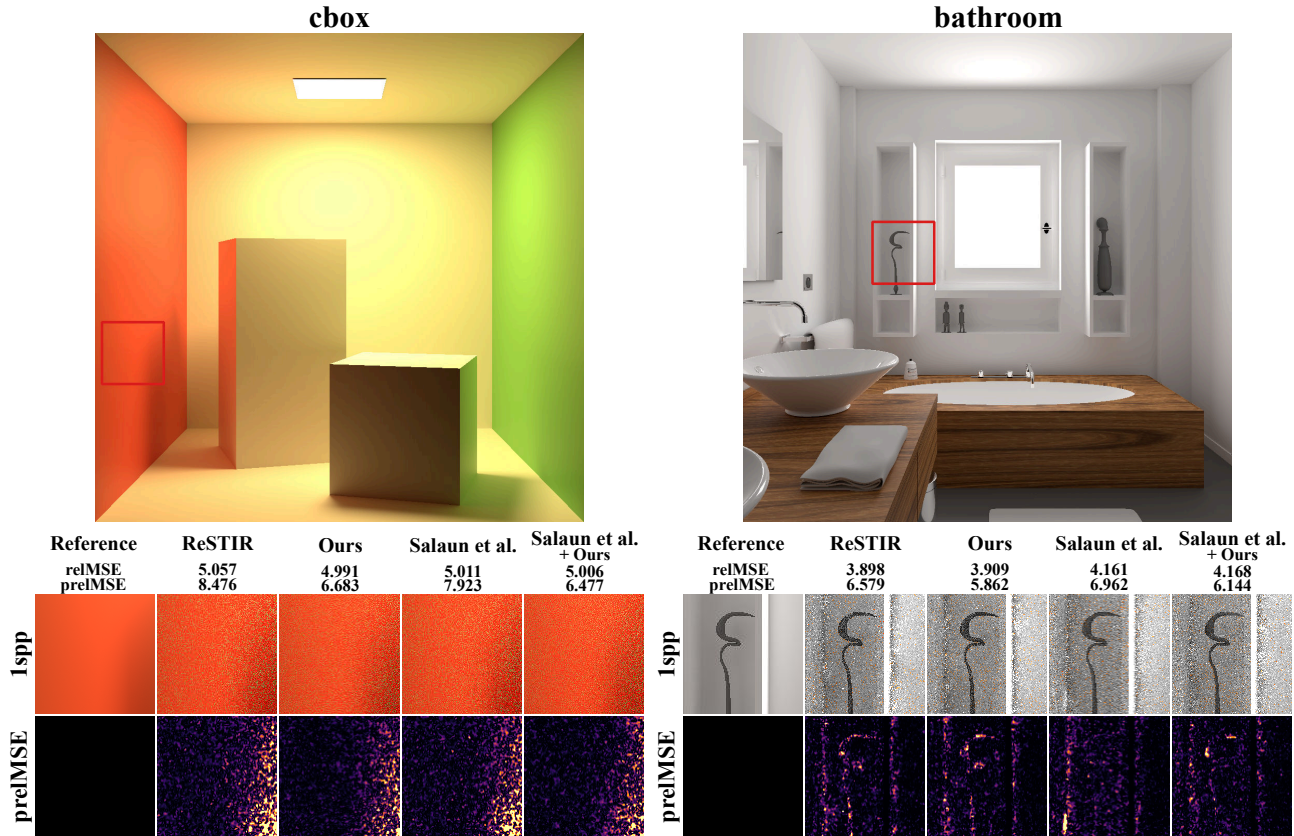


Fig. 7. Visual comparison of baseline ReSTIR, histogram stratification, BlueReSTIR, and histogram stratification combined with our resampling strategy under the unbiased implementation.

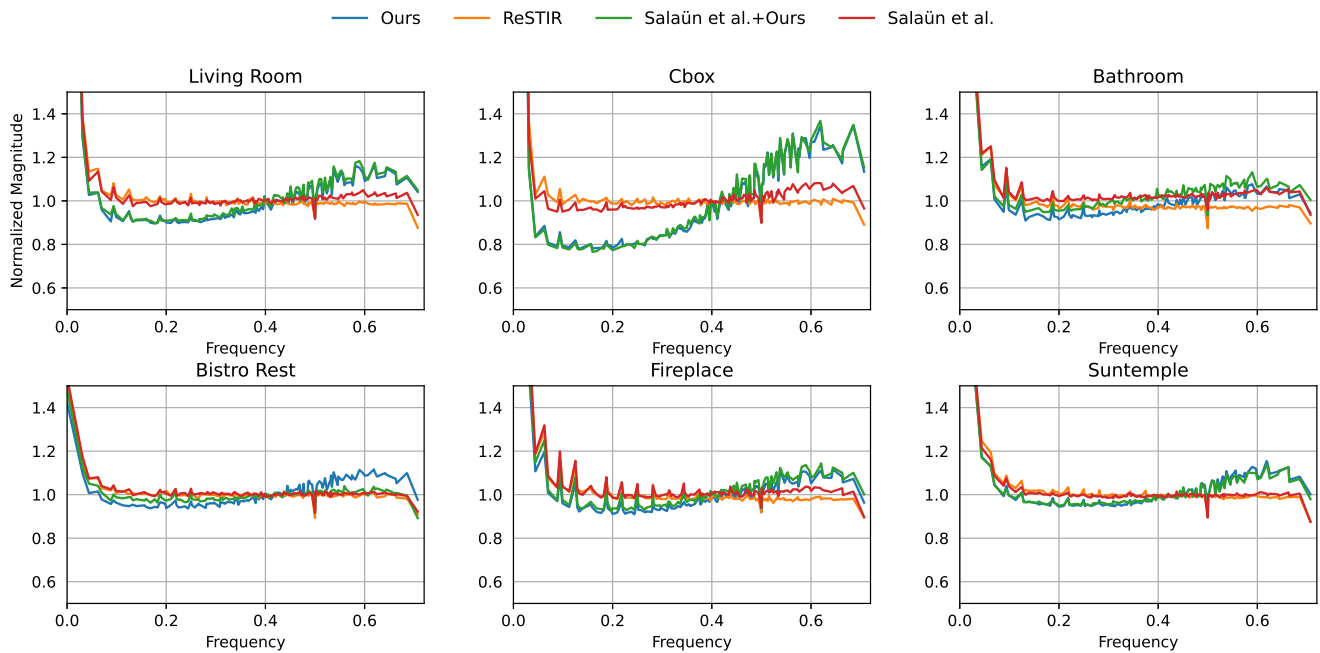


Fig. 8. Spectrum-energy distributions of the residual noise for baseline ReSTIR, histogram stratification from Salaun et al. [31], BlueReSTIR (Ours), and their combination (Salaun et al. + Ours) under the unbiased implementation.

TABLE III
NEIGHBOR-COUNT ABLATION USING PRELMSE.

Scene	Variant	$n = 3$	$n = 4$	$n = 5$	$n = 6$
cbox	blue	0.0072	0.0070	0.0067	0.0066
	ind	0.0084	0.0083	0.0085	0.0084
living_room	blue	0.0032	0.0032	0.0031	0.0031
	ind	0.0036	0.0036	0.0036	0.0036
suntemple	blue	0.0131	0.0125	0.0123	0.0121
	ind	0.0136	0.0133	0.0131	0.0129

that the perceptual benefit of our resampling strategy does not depend on a narrowly tuned neighborhood size. As the number of reused neighbors increases from 3 to 6, both methods generally improve slightly because each pixel receives a richer candidate set. However, BlueReSTIR maintains a measurable advantage throughout this range by shaping the final resampling decision rather than relying only on the amount of reuse.

The trend is especially visible in *CBox* and *Sun Temple*, where the gain of BlueReSTIR becomes slightly larger as more neighbors are included. This suggests that our method benefits from the increased candidate diversity and can convert it into a more favorable spatial error distribution. In *Living Room*, the improvement is more stable across different neighbor counts, indicating that BlueReSTIR remains effective even when the underlying candidate statistics vary less strongly. Overall, these results show that the method is robust with respect to the spatial-neighbor parameter and does not require a special operating point to reduce prelmse in these tests.

F. Denoising Evaluation

As a small additional experiment, we also examine how the residual error behaves under a standard denoiser. Since denoising acts as a low-pass filter, this serves as an experimental demonstration of blue noise's ability to remain favorable after filtering. Figure 9 compares denoised Cornell Box images using OIDN. In this setting, BlueReSTIR can still produce slightly cleaner edges and fewer residual blotches, which is consistent with its lower low-frequency error. Table I reports denoised results for all scenes. The denoised DSSIM values are generally close, with BlueReSTIR lower in most cases. This suggests that the blue-noise error shaping remains beneficial after denoising, although the effect is scene-dependent and more modest than the pre-denoising visual difference. The Sun Temple scene is a minor exception, where complex geometry reduces the visible benefit.

V. CONCLUSION

We have explored the intrinsic connection between dithering and resampling in ReSTIR, leading to a novel dithered resampling strategy that is fully compatible with existing ReSTIR pipelines. We introduce an efficient resampling algorithm, Quick Dithered Resample (QDR), which enables practical blue-noise-driven sampling for real-time rendering. The proposed method provides a complete integration of blue-noise dithering into ReSTIR's spatial resampling process, yielding improved perceptual quality in our experiments and more

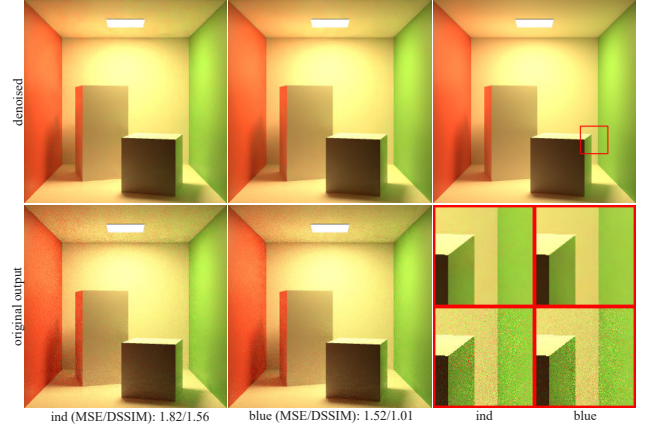


Fig. 9. The Cornell Box (CBox) before and after denoising. Left: baseline ReSTIR (ind). Right: BlueReSTIR (blue). Numbers indicate MSE ($\times 10^{-4}$) and DSSIM ($\times 10^{-2}$) (lower for better similarity). This scene serves as an experimental illustration of how the different residual error distributions behave after denoising.

desirable error spectra. The resulting high-frequency noise distribution supports direct visualization and remains compatible with denoising. The proposed method can be seamlessly integrated into existing ReSTIR renderers with minimal code modification and small runtime overhead.

Nevertheless, the current integration is still limited by the correlation between the blue-noise mask and the final resampling outcome. When the number of candidate samples is limited, the histogram used for CDF sampling becomes a coarse approximation of the complete sample distribution, which weakens the blue-noise properties. This limitation could potentially be mitigated through advanced dithering or perceptual optimization techniques such as those proposed by Chizhov et al. [26], though these approaches would introduce additional computational cost. We also observe that when ReSTIR employs highly accurate intensity estimations (as in ReSTIR PT), most residual noise manifests in color channels rather than intensity, reducing the perceptual advantage of blue-noise shaping, since human sensitivity is higher for luminance variation.

We believe this work bridges a conceptual gap between sampling theory and perceptual error control, providing a simple yet effective step toward perceptually guided real-time path tracing. Beyond real-time rendering, the proposed framework provides a general perspective linking stochastic quantization and Monte Carlo resampling, which could inspire more quasi-Monte Carlo strategies for light transport estimation.

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REFERENCES

- [1] B. Bitterli, C. Wyman, M. Pharr, P. Shirley, A. Lefohn, and W. Jarosz, "Spatiotemporal reservoir resampling for real-time ray tracing with dynamic direct lighting," *ACM Trans. Graph.*, vol. 39, no. 4, Aug. 2020. [Online]. Available: <https://doi.org/10.1145/3386569.3392481>

- [2] D. Lin, M. Kettunen, B. Bitterli, J. Pantaleoni, C. Yuksel, and C. Wyman, "Generalized resampled importance sampling: foundations of restir," *ACM Trans. Graph.*, vol. 41, no. 4, Jul. 2022. [Online]. Available: <https://doi.org/10.1145/3528223.3530158>
- [3] C. Wyman, M. Kettunen, D. Lin, B. Bitterli, C. Yuksel, W. Jarosz, and P. Kozłowski, "A gentle introduction to restir path reuse in real-time," in *ACM SIGGRAPH 2023 Courses*, ser. SIGGRAPH '23. New York, NY, USA: Association for Computing Machinery, 2023. [Online]. Available: <https://doi.org/10.1145/3587423.3595511>
- [4] R. Sawhney, D. Lin, M. Kettunen, B. Bitterli, R. Ramamoorthi, C. Wyman, and M. Pharr, "Decorrelating restir samplers via mcmc mutations," *ACM Trans. Graph.*, vol. 43, no. 1, Jan. 2024. [Online]. Available: <https://doi.org/10.1145/3629166>
- [5] S. Zhang, D. Lin, M. Kettunen, C. Yuksel, and C. Wyman, "Area restir: Resampling for real-time defocus and antialiasing," *ACM Trans. Graph.*, vol. 43, no. 4, Jul. 2024. [Online]. Available: <https://doi.org/10.1145/3658210>
- [6] R. Ulichney, *Digital halftoning*. Cambridge, MA, USA: MIT Press, 1987.
- [7] E. Cikalak, A. Gruson, I. Georgiev, D. Nowrouzezahrai, and T. Hachisuka, "Single-pass stratified importance resampling," in *Computer Graphics Forum*, vol. 41, no. 4. Wiley Online Library, 2022, pp. 41–49.
- [8] S. Kallweit, P. Clarberg, C. Kolb, T. Davidović, K.-H. Yao, T. Foley, Y. He, L. Wu, L. Chen, T. Akenine-Möller, C. Wyman, C. Crassin, and N. Benty, "The Falcor rendering framework," 8 2022, <https://github.com/NVIDIAGameWorks/Falcor>. [Online]. Available: <https://github.com/NVIDIAGameWorks/Falcor>
- [9] L. Schuchman, "Dither signals and their effect on quantization noise," *IEEE Transactions on Communication Technology*, vol. 12, no. 4, pp. 162–165, 1964.
- [10] J. B. Rodriguez, G. R. Arce, and D. L. Lau, "Blue-noise multitone dithering," *IEEE Transactions on Image Processing*, vol. 17, no. 8, pp. 1368–1382, 2008.
- [11] B. Bayers, "An optimum method for two level rendition of continuous tone pictures," in *Proc. of IEEE Int. Communication Conf.*, 1973, pp. 2611–2615.
- [12] R. A. Ulichney, "Void-and-cluster method for dither array generation," in *Human Vision, Visual Processing, and Digital Display IV*, vol. 1913. SPIE, 1993, pp. 332–343.
- [13] R. W. Floyd, "An adaptive algorithm for spatial gray-scale," in *Proc. Soc. Inf. Disp.*, vol. 17, 1976, pp. 75–77.
- [14] M. Analoui and J. P. Allebach, "Model-based halftoning using direct binary search," in *Human vision, visual processing, and digital display III*, vol. 1666. SPIE, 1992, pp. 96–108.
- [15] I. Georgiev and M. Fajardo, "Blue-noise dithered sampling," in *ACM SIGGRAPH 2016 Talks*, ser. SIGGRAPH '16. New York, NY, USA: Association for Computing Machinery, 2016. [Online]. Available: <https://doi.org/10.1145/2897839.2927430>
- [16] W. Donnelly, A. Wolfe, J. Büttepage, and J. Valdés, "Fast: Filter-adapted spatio-temporal sampling for real-time rendering," *Proc. ACM Comput. Graph. Interact. Tech.*, vol. 7, no. 1, May 2024. [Online]. Available: <https://doi.org/10.1145/3651283>
- [17] A. Wolfe, N. Morrical, T. Akenine-Möller, and R. Ramamoorthi, "Spatiotemporal Blue Noise Masks," in *Eurographics Symposium on Rendering*, A. Ghosh and L.-Y. Wei, Eds. The Eurographics Association, 2022.
- [18] E. Heitz, L. Belcour, V. Ostromoukhov, D. Coeurjolly, and J.-C. Iehl, "A low-discrepancy sampler that distributes monte carlo errors as a blue noise in screen space," in *ACM SIGGRAPH 2019 Talks*, ser. SIGGRAPH '19. New York, NY, USA: Association for Computing Machinery, 2019. [Online]. Available: <https://doi.org/10.1145/3306307.3328191>
- [19] A. G. M. Ahmed and P. Wonka, "Screen-space blue-noise diffusion of monte carlo sampling error via hierarchical ordering of pixels," *ACM Trans. Graph.*, vol. 39, no. 6, Nov. 2020. [Online]. Available: <https://doi.org/10.1145/3414685.3417881>
- [20] A. G. M. Ahmed, J. Ren, and P. Wonka, "Gaussian blue noise," *ACM Trans. Graph.*, vol. 41, no. 6, Nov. 2022. [Online]. Available: <https://doi.org/10.1145/3550454.3555519>
- [21] F. de Goes, K. Breeden, V. Ostromoukhov, and M. Desbrun, "Blue noise through optimal transport," *ACM Trans. Graph.*, vol. 31, no. 6, Nov. 2012. [Online]. Available: <https://doi.org/10.1145/2366145.2366190>
- [22] C. Salatin, I. Georgiev, H.-P. Seidel, and G. Singh, "Scalable multi-class sampling via filtered sliced optimal transport," *ACM Trans. Graph.*, vol. 41, no. 6, Nov. 2022. [Online]. Available: <https://doi.org/10.1145/3550454.3555484>
- [23] A. G. M. Ahmed, J. Guo, D.-M. Yan, J.-Y. Franceschia, X. Zhang, and O. Deussen, "A simple push-pull algorithm for blue-noise sampling," *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 12, pp. 2496–2508, 2017.
- [24] Z. Chen, Z. Yuan, Y.-K. Choi, L. Liu, and W. Wang, "Variational blue noise sampling," *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 10, pp. 1784–1796, 2012.
- [25] E. Heitz and L. Belcour, "Distributing monte carlo errors as a blue noise in screen space by permuting pixel seeds between frames," in *Computer Graphics Forum*, vol. 38, no. 4. Wiley Online Library, 2019, pp. 149–158.
- [26] V. Chizhov, I. Georgiev, K. Myszkowski, and G. Singh, "Perceptual error optimization for monte carlo rendering," *ACM Trans. Graph.*, vol. 41, no. 3, Mar. 2022. [Online]. Available: <https://doi.org/10.1145/3504002>
- [27] M. Korać, C. Salaün, I. Georgiev, P. Grittmann, P. Slusallek, K. Myszkowski, and G. Singh, "Perceptual error optimization for monte carlo animation rendering," in *SIGGRAPH Asia 2023 Conference Papers*, ser. SA '23. New York, NY, USA: Association for Computing Machinery, 2023. [Online]. Available: <https://doi.org/10.1145/3610548.3618146>
- [28] J. F. Talbot, *Importance resampling for global illumination*. Brigham Young University, 2005.
- [29] Y. Ouyang, S. Liu, M. Kettunen, M. Pharr, and J. Pantaleoni, "Restir gi: Path resampling for real-time path tracing," in *Computer Graphics Forum*, vol. 40, no. 8. Wiley Online Library, 2021, pp. 17–29.
- [30] J. Liu, D. Lin, M. Kettunen, C. Wyman, and R. Ramamoorthi, "Reservoir splatting for temporal path resampling and motion blur," in *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*, ser. SIGGRAPH Conference Papers '25. New York, NY, USA: Association for Computing Machinery, 2025. [Online]. Available: <https://doi.org/10.1145/3721238.3730646>
- [31] C. Salaün, M. Balint, L. Belcour, E. Heitz, G. Singh, and K. Myszkowski, "Histogram stratification for spatio-temporal reservoir sampling," in *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*, 2025, pp. 1–10.
- [32] A. T. Áfra, "Intel® Open Image Denoise," 2025, <https://www.openimagedenoise.org>.
- [33] M. McGuire, "Computer graphics archive," July 2017. [Online]. Available: <https://casual-effects.com/data>
- [34] A. Lumberyard, "Amazon lumberyard bistro, open research content archive (orca)," July 2017. [Online]. Available: <http://developer.nvidia.com/orca/amazon-lumberyard-bistro>
- [35] E. Games, "Unreal engine sun temple, open research content archive (orca)," October 2017. [Online]. Available: <http://developer.nvidia.com/orca/epic-games-sun-temple>



Zhong Shi is a graduate student in the Department of Computer Science and Technology, Tsinghua University, Beijing, China. He received the bachelor's degree in Computer Science and Technology from Tsinghua University in 2024. His research interests include Monte Carlo path tracing and sampling.



Kun Xu is an associate professor in the Department of Computer Science and Technology, Tsinghua University. Before that, he received his bachelor and doctor degrees from the same university in 2005 and 2009, respectively. His research interests include realistic rendering and image/video editing.